Stochastic Modeling and Simulation
3rd Y - Semester 5
2025-2026

The National School of Artificial Intelligence
المدرسة الوطنية العليا للذكاء الاصطناعي

Answers Key Midterm Exam

Exercise 1 *Let X_n denote "the number of failed machines at the start of day n".*
Scenario 1: *The state space is $S = \{0, 1\}$ and*

1. *Since the number of machines that are down in the morning depends only on the number that were down the previous morning, independently of all other days, the process indeed forms a Markov chain; so we have:*

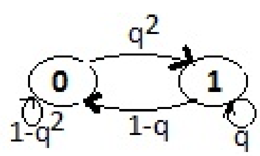
   $p_{00} = \mathbb{P}(\textit{no failed during the day or 1 failed}) = (1-q)^2 + 2q(1-q) = 1 - q^2$

   $p_{01} = \mathbb{P}(\textit{2 failures during the day}) = q^2$

   $p_{10} = \mathbb{P}(\textit{the failed machine is repaired and the other one is working}) = 1 - q$

   $p_{11} = \mathbb{P}(\textit{the failed machine is repaired and the other one is failed}) = q$

   Transition matrix and graph are as follows:



$$P = \begin{pmatrix} 1-q^2 & q^2 \\ 1-q & q \end{pmatrix}$$

(1,5)

2. *Stationary distribution: Let $\pi = (\pi_0, \pi_1)$, for $0 < q < 1$ the chain is irreducible and apperidic, then it satisfy $\pi = \pi P$ and $\pi_0 + \pi_1 = 1.$ We get*

$$\pi_1 = \frac{q^2}{1-q}\pi_0$$

so

$$\pi_0 = \frac{1-q}{1-q+q^2} \textit{ and } \pi_1 = \frac{q^2}{1-q+q^2}$$

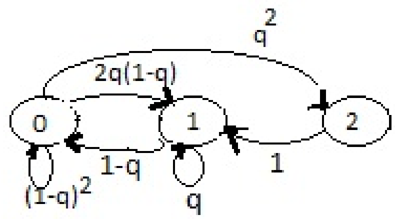
Numerical application: for $q = \frac{1}{4}$ we obtain $\pi_0 = \frac{12}{13}$ and $\pi_1 = \frac{1}{13}$.

(1,5)

Scenario 2: *Here, both machines can be broken at the start of a day, because no machines are repaired the night before, so the state space is $S = \{0, 1, 2\}$ and*

$$\begin{aligned} p_{00} &= (1-q)^2; \quad p_{01} = 2q(1-q); \quad p_{02} = q^2; \\ p_{10} &= 1-q; \qquad p_{11} = q; \qquad p_{12} = 0; \\ p_{21} &= 1. \end{aligned}$$

1. *Transition matrix and graph:*



$$P = \begin{pmatrix} (1-q)^2 & 2q(1-q) & q^2 \\ (1-q) & q & 0 \\ 0 & 1 & 0 \end{pmatrix} \qquad (1{,}5)$$

For $0 < q < 1$ *all states communicate then the chain is irreducible and apperiodic since* $d(0) = 1$.

2. *Stationary distribution: Let* $\pi = (\pi_0, \pi_1, \pi_2)$, *the chain is ergodic, then it satisfy* $\pi = \pi P$ *and* $\pi_0 + \pi_1 + \pi_2 = 1$. *We getso*

$$\pi_0 = \frac{(1-q)(1-q+q^2)}{1-q+q^2-q^3+q^4},\ \pi_1 = \frac{q(1-q)^2(2q)}{1-q+q^2-q^3+q^4} \text{ and } \pi_2 = \frac{q^2}{1-q+q^2-q^3+q^4}$$

Numerical application: for $q = \frac{1}{4}$ *we obtain*

$$\pi = (\pi_0, \pi_1, \pi_2) = \left(\frac{48}{79}, \frac{28}{79}, \frac{3}{79}\right). \qquad (1{,}5)$$

3. *Expected time to total failure: we have* $m_{i2} = E[T_2/X_0 = i] = E_i[T_2]$

We have $E_2[T_2] = 0$, *and the following system equations:*

$$\begin{cases} E_0[T_2] = 1 + (1-q)^2 E_0[T_2] + 2q(1-q) E_1[T_2] + q^2 E_2[T_2] \\ E_1[T_2] = 1 + (1-q) E_0[T_2] + q E_1[T_2] \end{cases}$$

so, we get

$$E_0[T_2] = \frac{2q+1}{q^2} \text{ and } E_1[T_2] = \frac{q^2-q-1}{q^2(q-1)}$$

(1,5)

hence for $q = \frac{1}{4}$, $E_0[T_2] = 24$ *days and* $E_1[T_2] = 25.33$ *days.*

Then, starting from a fully functional system, it takes on average 24 *days for both machines to be down simultaneously.*

4. *Expected time spent in each state before returning to* 0 *is*

$$m_0 = \frac{1}{\pi_0} = \frac{79}{48} \simeq 1.65 days. \qquad (0{,}5)$$

5. *Average daily failure rate: If we are in state i, so there are i machines failures, then the number of working machines in the journey is $(2-i)$, hence*

$$E\left[new\ failures/X_i = i\right] = (2-i)\,q$$

then

$$\begin{aligned} E\left[new\ failures\right] &= \sum_i \pi_i\,(2-i)\,q = q\sum_i \pi_i\,(2-i) \\ E\left[new\ failures\right] &= q\,(2\pi_0+\pi_1) \end{aligned} \quad (0,5)$$

Scenario 1: *for $q=\frac{1}{4}$, we get*

$$E\left[new\ failures\right] = \frac{1}{4}\left(2\times\frac{12}{13}+\frac{1}{13}\right) = \frac{25}{52} = 0.48 \quad (0,5)$$

Scenario 2: *for $q=\frac{1}{4}$, we get*

$$E\left[new\ failures\right] = \frac{1}{4}\left(2\times\frac{48}{79}+\frac{28}{79}\right) = \frac{31}{79} = 0.39 \quad (0,5)$$

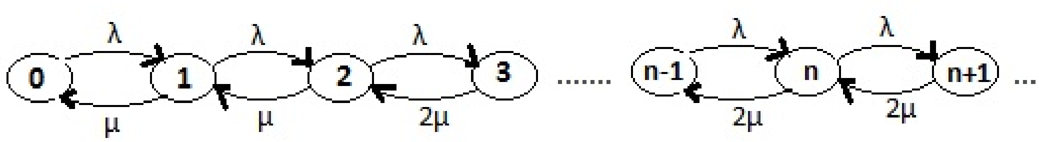
Interpretation: *thus, more frequent maintenance in Scenario 1 ensures higher uptime, even though more failures occur per day statistically.* (0,5)

Exercise 2 1. *We are in the case of queueing Model with variable service. The state of the system can be described by the stochastic process $\{N(t), t\geq 0\}$ that is a birth and death process, such that $N(t)$ is number of cars in customs post with state space $S=\{0,1,2,\cdots\}$ and transition rates are*

$$\lambda_n = \lambda,\ \ for\ n\geq 0\ \ and\ \mu_n = \begin{cases} 0\ for\ n=0 \\ \mu\ for\ n=1\ \ or\ 2 \\ 2\mu\ for\ n\geq 3 \end{cases}. \quad (1)$$

For n large the service rate is equal to 2μ, then the condition of the stability of the system is $\lambda < 2\mu$, or the total traffic intensity $\rho = \frac{\lambda}{2\mu} < 1$. (0,5)

Then the simplified transition graph is given as follows



(0,5)

2. *In Transitary State of the system, let $p_n(t) = P(N(t)=n), n\geq 0$. From the graph we obtain the following Kolmogorov equations system:*

$$\begin{cases} p_0'(t) = -\lambda p_0(t) + \mu p_1(t) \\ p_1'(t) = \lambda p_0(t) - (\lambda+\mu)\,p_1(t) + \mu p_2(t) \\ p_2'(t) = \lambda p_1(t) - (\lambda+\mu)\,p_2(t) + 2\mu p_3(t) \\ p_n'(t) = \lambda p_{n-1}(t) - (\lambda+2\mu)\,p_n(t) + 2\mu p_{n+1}(t),\, n\geq 3 \end{cases} \quad (1,5)$$

In Stationary State of the system, let $\lim_{t\to\infty} p_n(t) = p_n, n \geq 0$ *exist, then from the Kolmogorov equations we obtain the following balance equations system:*

$$\left\{\begin{array}{c} 0 = -\lambda p_0 + \mu_1 p_1 \\ 0 = \lambda p_0 - (\lambda+\mu)\, p_1 + \mu p_2 \\ 0 = \lambda p_1 - (\lambda+\mu)\, p_2 + 2\mu p_3 \\ 0 = \lambda p_{n-1} - (\lambda+2\mu)\, p_n + 2\mu p_{n+1}, n \geq 3 \end{array}\right. \qquad (1.0)$$

with $\sum_{n=0}^{\infty} p_n = 1$. *Solving these equations gives us the stationary distribution of the number of customers in the system:*

$$\begin{aligned} p_1 &= \frac{\lambda}{\mu} p_0, \\ p_2 &= \left(\frac{\lambda}{\mu}\right)^2 p_0 \\ p_n &= \left(\frac{\lambda}{\mu}\right)^2 \left(\frac{\lambda}{2\mu}\right)^{n-2} p_0, \text{ for } n \geq 3. \end{aligned} \qquad (0{,}5)$$

Calculating p_0 *using* $\sum_{n=0}^{\infty} p_n = 1$:

$$p_0 = \left[1 + \frac{\lambda}{\mu} + \left(\frac{\lambda}{\mu}\right)^2 \left(1 + \frac{\lambda}{2\mu} + \left(\frac{\lambda}{2\mu}\right)^2 + \ldots\right)\right]^{-1}$$

setting $\alpha = \frac{\lambda}{\mu}$,

$$p_0 = \left[1 + \alpha + \alpha^2 \left(\frac{1}{2-\alpha/2}\right)\right]^{-1} = \frac{2-\alpha}{2+\alpha+\alpha^2}$$

then

$$p_n = \left\{\begin{array}{l} \alpha^n p_0, \; n = 0,1,2 \\ \alpha^2 \rho^{n-2} p_0, \; n \geq 3 \end{array}\right. \qquad (0{,}25)$$

3. *Numerical application: given that* $\lambda = 10$ *cars/hour and* $\frac{1}{\mu} = 5$ *minutes* $\Rightarrow \mu = 12$ *cars/minutes, so* $\alpha = \frac{10}{12} = \frac{5}{6}$ *and* $\rho = \frac{5}{12}$; *then*

$$p_0 = \frac{2-\alpha}{2+\alpha+\alpha^2} = \frac{42}{127} = 0.3307 \qquad (0{,}25)$$

a) *The average number of cars in the queue:*

$$\begin{aligned} \overline{n_q} &= E[N_q] = p_2 + \sum_{n=3}^{+\infty} (n-2)\, p_n = \alpha^2 p_0 + \alpha^2 p_0 \sum_{n=3}^{+\infty} (n-2)\, \rho^{n-2} \\ &= \alpha^2 p_0 \left[1 + \rho \sum_{n=3}^{+\infty} (n-2)\, \rho^{n-3}\right] = \alpha^2 p_0 \left[1 + \frac{\rho}{(1-\rho)^2}\right] \\ &= \left(\frac{5}{6}\right)^2 \frac{42}{127} \left[1 + \frac{\frac{5}{12}}{(1-\frac{5}{12})^2}\right] = \frac{5^2}{6} \frac{1}{127} \frac{109}{7} \\ \overline{n_q} &= 0.5109 \end{aligned} \qquad (1{,}5)$$

b) *Calculate the average waiting time for a car.*

$$\overline{W} = \frac{\overline{n_q}}{\lambda} = 0.5109 \times \frac{1}{10} = 0.05109h \simeq 3\,\text{min}\,4\,\text{sec} \qquad \text{(0,5)}$$

c) *Determine the average time required to check a car:* $\overline{W_{_{ser}}} = \overline{W_s} - \overline{W}$, *or* $\overline{W_{_{ser}}} = \frac{1}{\mu}$ *we have so*

$$\overline{W_{_{ser}}} = \frac{1}{\mu} = 5\,\text{min} \qquad \text{(0,5)}$$

d) *Calculate the probability that:*

Exactly one officer busy,

$$P\left(1\ \ busy\right) = p_1 + p_2 = p_0\left(\alpha + \alpha^2\right) = \frac{385}{762} = 0.5052 \qquad \text{(0,5)}$$

Both officers busy,

$$P\left(2\ \ busy\right) = \sum_{n=3}^{+\infty} p_n = \alpha^2 p_0 \sum_{n=3}^{+\infty} \rho^{n-2} = \alpha^2 p_0 \frac{\rho}{(1-\rho)} = \frac{125}{762} = 0.1640 \qquad \text{(0,5)}$$

50.52% *of time there is only one officer is busy and about* 16.4% *of time both officers are busy. Thus, the customs post is stable and efficient under the given rates.*

(0,5)